

Rotational Speed Measurement & Statistical Data Analysis

Comparing five methods for measuring motor speed and analyzing measurement variability

Author	Luis Ramirez	Performed	Spring 2026
Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
Team	Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortez, Abhin Ghandi, Ramid Brown	Equipment	DC motor rig, digital tachometer, optical sensor, frequency counter, stroboscope, oscilloscope, signal generator, multimeter

PROJECT SUMMARY

This experiment measured the rotational speed of a DC motor using several independent techniques and examined how speed relates to frequency and voltage. A DC generator output was calibrated against motor speed using least-squares regression, giving a nearly perfect linear relationship of about 10 mV per rpm ($R^2 = 0.99995$). A digital contact tachometer was used by all five team members to collect 75 readings, which averaged 995.9 rpm with a standard deviation of 2.13 rpm. Stroboscope synchronization and oscilloscope Lissajous figures were used to confirm the speed optically and electrically. Comparing the methods showed that the electrically based regression was the most consistent, while contact and visual methods introduced operator-dependent variability.

1. Introduction & Objectives

The objective was to learn how to measure rotational speed and how frequency relates to revolutions per minute (rpm). A disk indexed at equal intervals was monitored and the frequency at which the intervals passed a fixed sensor was counted; with evenly spaced slots, this frequency is directly proportional to rpm. Voltage-based measurements were also calibrated so that speed could be read electrically, and statistical tools were used to evaluate the tachometer data.

2. Equipment

Table 1. Instruments used

Instrument	Role in the experiment
DC motor rig	Provides the rotating shaft, with a tachometer point, a magnetic pickup, and a DC generator output that produces a speed-dependent voltage
Frequency counter	Counts full signal cycles per second (Hz), filtering noise for an accurate frequency reading
Digital contact tachometer	Pressed against the shaft center with a rubber tip to read speed directly in rpm
Signal generator	Produces a controlled sine, square, or sawtooth signal of known frequency for comparison
Digital multimeter	Measures the DC generator output voltage
Oscilloscope	Displays signals and overlays two inputs to form Lissajous figures
Stroboscope	Flashes light at a set rate; when the rate matches the shaft speed, a reference mark appears frozen

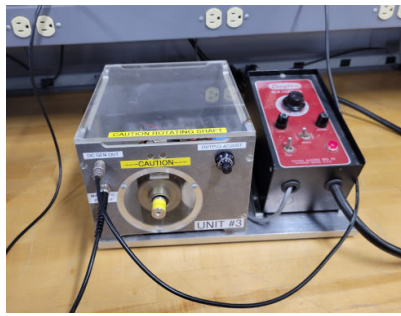


Figure 1. DC motor rig



Figure 2. Sensor / voltage measurement setup

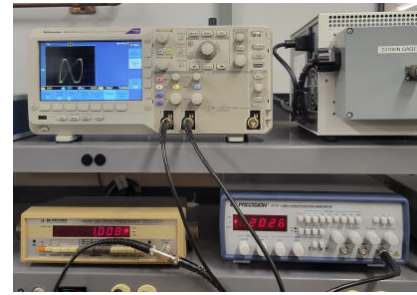


Figure 3. Oscilloscope and signal generator setup

3. Results

Least-squares calibration: speed vs. voltage

Table 2. Speed, frequency, and voltage readings

Motor speed (rpm)	Counter frequency (Hz)	Generator voltage (V)
1500	1495	14.9923
1200	1200	11.9879
900	893	8.9591
600	600	6.0681
300	300	2.9629
0	0	0

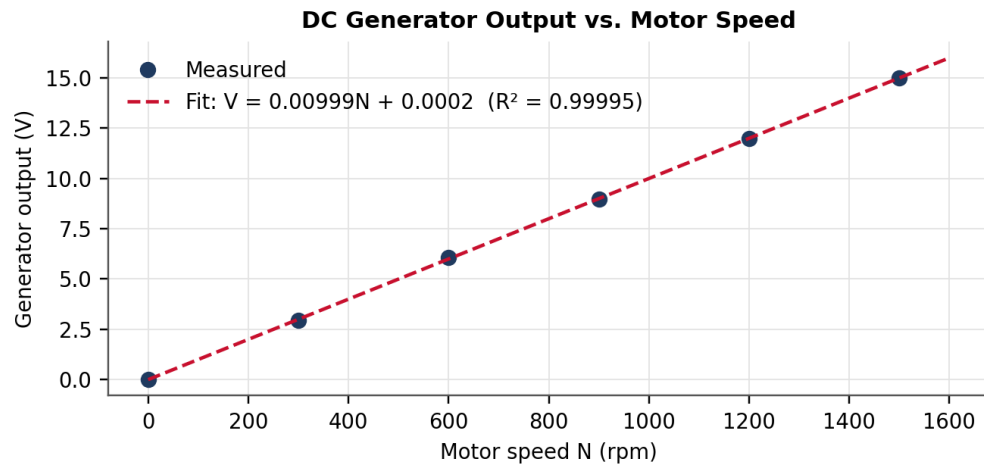


Figure 4. Generator output voltage vs. motor speed with least-squares fit.

The regression gives $V = 0.00999N + 0.0002$, a sensitivity of almost exactly 10 mV per rpm with essentially zero offset. The frequency counter tracked the set speed within 0.8% at every point.

Tachometer repeatability

Each of the five team members took 15 tachometer readings with the motor at a constant speed, for 75 readings in total.

Table 3. Descriptive statistics of tachometer readings

Statistic	Value	Statistic	Value
Count	75	Range	989.6 – 1004 rpm
Mean	995.85 rpm	Median	995.6 rpm
Standard deviation	2.13 rpm	Mode	995.5 rpm
Variance	4.54 rpm ²	Skewness	0.38
Coefficient of variation	0.21%	Excess kurtosis	3.62

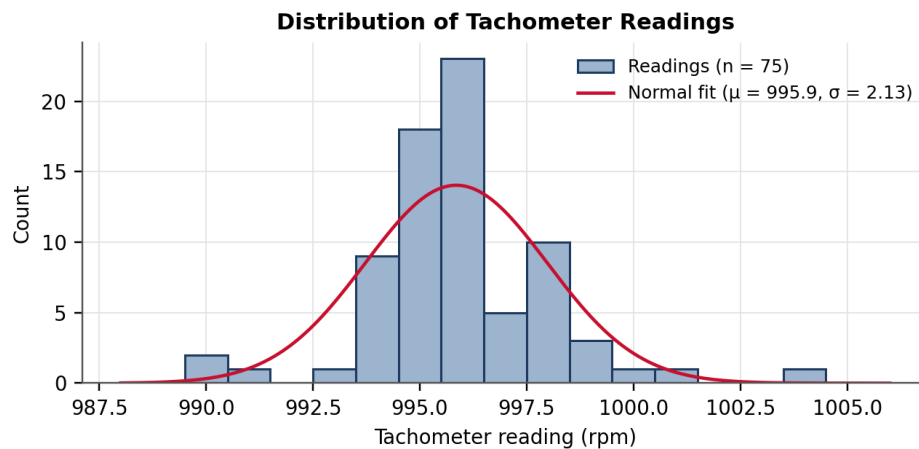


Figure 5. Histogram of all 75 tachometer readings with a fitted normal curve.

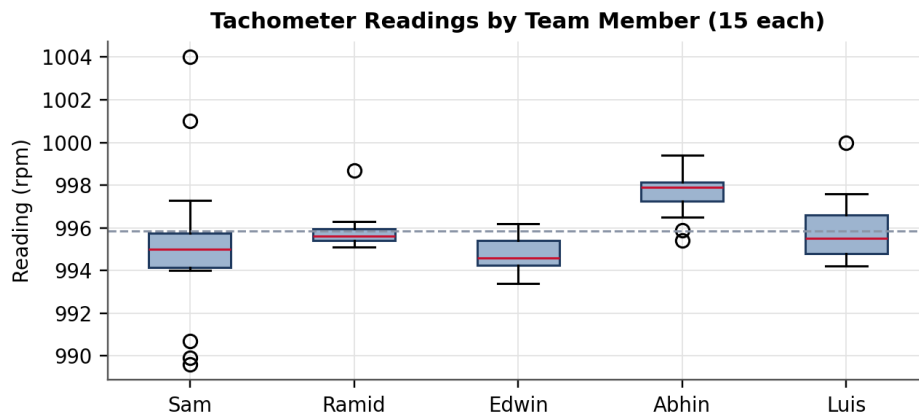


Figure 6. Readings grouped by team member.

Stroboscope and Lissajous figures

Table 4. Stroboscope patterns

Strobe rate	Pattern observed
997.2	1 stationary line (flash rate = shaft speed)
1992	2 lines (twice the shaft speed)
3000	3 lines (three times the shaft speed)
500	Partial line (half the shaft speed)

The single stationary line at a strobe rate of 997.2 agrees with the tachometer mean of about 996 rpm. On the oscilloscope, the motor signal was compared with the signal generator: matched frequencies produced a closed loop, and 2:1 and 1:2 frequency ratios produced figure-eight patterns. The generator frequencies that gave the clearest patterns differed slightly from the exact $1\times$, $2\times$, and $0.5\times$ ratios, mainly because judging the most stable pattern is subjective.



Figure 7. 1:1 frequency ratio



Figure 8. Generator at half frequency



Figure 9. Generator at twice the frequency

4. Discussion

Using several independent methods allowed each one to cross-check the others. The **least-squares method** produced the most consistent numbers because it relied on electrical output and regression rather than mechanical contact or visual judgment. The calibration was essentially perfectly linear, which is exactly what is expected from a DC generator whose voltage is proportional to speed.

The **tachometer** data were tightly grouped: the standard deviation of 2.13 rpm is only 0.21% of the mean. The histogram is centered near 996 rpm but has heavier tails than a normal distribution (excess kurtosis 3.6), and the box plot shows why — a handful of readings from one team member (989.6 to 1004 rpm) are far from the rest, while the other four members had standard deviations of about 1 rpm or less. One member's average was also about 1.8 rpm above the group mean, a small systematic difference. This is typical of contact tachometers, where tip pressure and alignment change the reading.

The **stroboscope** confirmed the speed optically and made harmonic behavior visible: matching the shaft speed froze one line, while multiples and fractions of the speed produced multiple or partial lines. The **Lissajous** method confirmed frequency ratios electrically, but both of these visual methods depend on the observer's judgment.

5. Conclusion

The experiment measured motor speed through mechanical contact, optical synchronization, electrical signal analysis, and mathematical modeling. All methods agreed on a speed of roughly 996 rpm, and the least-squares calibration was the most numerically consistent because it removed operator-dependent variation. The spread in the tachometer data showed how much human technique influences a measurement, reinforcing that engineering accuracy is not just about obtaining ideal numbers but about understanding where the numbers come from.

6. Key Concepts

How does a stroboscope indicate whether it is faster or slower than the shaft?

When the flash rate matches the shaft speed, the reference mark appears stationary. If the shaft turns slightly faster than the flash rate, the mark appears to creep forward in the direction of rotation; if it turns slightly slower, the mark appears to drift backward.

What is kurtosis?

A statistic describing how heavy a distribution's tails are compared with a normal distribution. High kurtosis means more extreme values than a normal distribution would predict; skewness, by contrast, describes asymmetry.

References

Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.

Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.